

SELF-SUPERVISED REPRESENTATION LEARNING FOR RAPID ADAPTATION UNDER CLINICAL CONCEPT DRIFT

Anonymous authors

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ABSTRACT

Under clinical concept drift from FDG to PSMA PET/CT, self-supervised pre-training on source-domain volumes yields representations that adapt to the target tracer with far fewer labeled cases than training from scratch—but this advantage is *shot-dependent* and collapses in the extreme low-shot / zero-shot regime unless the adaptation protocol is matched to the drift. We instantiate this claim with an aligned FDG→PSMA benchmark that freezes a shared FDG stage and varies only the PSMA label budget ($N \in \{50, 10, 5, 0\}$), reporting Dice together with voxel-level FP/FN. Across fifteen method variants, PET/CT MAE SSL leads at moderate budgets ($\text{fs}50/\text{fs}10$), yet the SSL-scratch gap shrinks to noise by $\text{fs}5/\text{fs}0$, where competing recipes can overtake—highlighting *when* representation learning does and does not buy robustness to clinical concept drift.¹

1 INTRODUCTION

We use *clinical concept drift* for tracer change in oncology PET/CT: not only covariate shift of images, but a change in what “lesion” looks like (uptake pattern, organ background) and how labels distribute (size, prevalence, empty-GT rate) when moving from FDG to PSMA (Chen et al., 2025). This differs from classical temporal concept drift; the inducing factor is a change of clinical measurement concept (tracer), not calendar time.

Self-supervised / foundation-style pretraining promises rapid adaptation when target labels are scarce (He et al., 2022; Tang et al., 2022; Anonymous, 2026), while nnU-Net-style supervised toolkits remain strong baselines (Isensee et al., 2021). Work on normative representation learning further stresses that clinical findings are rare and imperfectly observed, so average metrics on frequent phenotypes can mislead (Bercea et al., 2025b;a).

Core claim. Source-domain SSL can reduce the labeled PSMA budget needed for competitive segmentation under FDG→PSMA drift, but the gain is concentrated at *moderate* few-shot budgets and is not automatic at near-zero labels.

Testable sub-claims.

1. **Drift.** Cross-tracer lesion segmentation induces concept drift in lesion appearance and label statistics, beyond covariate shift alone.
2. **Representation helps.** Under a shared FDG→PSMA protocol, SSL initializations (e.g., MAE/MIM/dual-encoder) improve few-shot PSMA adaptation over scratch at moderate N (here, especially $N=50$).
3. **When it fails.** Gains shrink or reverse as labeled PSMA shots decrease; foundation-style or retrieval baselines may fail once the clinical concept has drifted.
4. **Evaluation.** Fair comparison requires an aligned protocol (same FDG stage → fs50/10/5/0) and metrics that separately report Dice, FP, and FN under long-tailed empty/positive cases.

¹ICLR workshop draft; numbers are live board snapshots.

Table 1: PSMA/FDG TEST **Dice (%)**. “–” = pending. **Bold** = best completed entry per column.

Method	Init	fs50	fs10	fs5	fs0	fc70	FDG
Proto+Retrieval	retrieval	0.0	0.0	0.0	0.0	0.2	0.0
nnU-Net MIM	MIM	41.6	36.9	32.5	19.1	44.0	61.2
nnU-Net	scratch	32.2	31.6	29.1	15.4	33.0	56.2
MONAI SwinViT	Tang SSL	44.6	29.4	26.1	15.5	56.1	61.0
MONAI SwinViT	scratch	46.1	32.9	29.2	11.5	57.5	61.0
DpDNet dual-enc	dual-enc	45.0	41.4	36.2	17.7	49.2	63.9
DpDNet	scratch	44.0	40.8	41.2	16.7	44.0	61.4
SegAnyPET	Lesion	15.6	8.4	8.0	26.0	22.0	44.7
SegAnyPET	scratch	8.0	3.7	3.0	22.1	19.2	35.6
PET/CT MAE	MAE SSL	53.0	39.3	30.6	16.0	60.5	73.2
PET/CT MAE	scratch	48.1	38.2	31.2	15.6	58.1	66.3
BIRTH	scratch	–	32.5	29.5	–	–	–
BIRTH	MT619	–	–	–	–	–	–
YixinChen	scratch	32.7	29.4	26.6	–	–	–
YixinChen	MT619	–	–	–	–	–	–

2 ALIGNED EXPERIMENTAL PROTOCOL

Data. We use a stratified PET/CT cohort with tracer-aware roles (PSMA/FDG $\times$ small/large/negative lesion burden) and a fixed 70/10/20 train/val/test split seed. PSMA few-shot subsets ($N = 50, 10, 5$) are drawn from the PSMA train pool; PSMA $fs0$ evaluates the shared FDG checkpoint zero-shot on PSMA TEST; $fc70\%$ uses 70% of PSMA train as a high-label reference; FDG TEST probes source-domain retention.

Shared FDG stage. Every non-retrieval method first trains (or fine-tunes) on the same FDG supervised split with matched global batch size and a common epoch budget policy for the board. The resulting FDG checkpoint initializes PSMA adaptation unless the method is training-free.

PSMA adaptation. Few-shot fine-tuning uses small labeled PSMA sets with online validation and TEST evaluation aggregated over nine TEST folds (means in the main text; per-fold grids omitted). Metrics:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}, \quad FP\% = \frac{FP}{Neg}, \quad FN\% = \frac{FN}{Pos},$$

with Dice excluding empty-GT cases and FP/FN as voxel micro-averages.

Methods. We compare scratch vs. pretrained siblings when available: PET/CT MAE SSL (Anonymous, 2026), Swin SSL (Tang et al., 2022), nnU-Net/MIM (Isensee et al., 2021), DpDNet (Anonymous, 2025a), SegAnyPET (Anonymous, 2025b), prototype retrieval, and AutoPET V entries (BIRTH / YixinChen).

3 RESULTS

Moderate few-shot: SSL buys rapid adaptation. At PSMA $fs50$, PET/CT MAE SSL reaches 53.0 Dice vs. 48.1 for its scratch twin (+4.9 pp) and leads completed methods (Table 1). The SSL edge remains positive but smaller at $fs10$ (+1.1 pp). This supports the “rapid adaptation” half of the claim under a matched FDG stage.

Extreme scarcity: the advantage collapses. By $fs5$, the MAE SSL–scratch gap flips slightly (−0.6 pp), and DpDNet scratch leads Dice (41.2). At $fs0$, most learned segmentors sit in a low band (~ 11 –19); SegAnyPET’s higher zero-shot Dice coexists with severe FN on few-shot columns. Retrieval does not transfer under this drift. Gains from SSL are therefore *shot-dependent*, not unconditional.

Failures are FN-dominated. Board FP rates are typically $\ll 1\%$, while FN often exceeds 30–70% as N decreases—Dice alone hides under-segmentation of sparse PSMA disease.

FDG → PSMA few-shot · mean TEST metrics (15 methods)
2026-09-04 09:55:35 CST · Dice / FP / FN (%) · no per-fold grids

Method (pretrained)	0 FDG (shared)	PSMA fs50	PSMA fs10	PSMA fs5	PSMA fs0	PSMA fs70%	FDG TEST
Proto-Retrieval (ECCV20) none (retrieval)	N/A	0.85% 0.78% 87.26%	0.85% 0.78% 87.26%	0.85% 0.78% 87.26%	0.85% 0.78% 87.26%	0.23% 17.75% 35.33%	0.85% 5.75% 75.99%
nnUNet MIM (Nat-Method v21) PET-CT MIM	DONE 100ep	41.57% 0.81% 45.74%	36.94% 0.96% 76.89%	32.45% 0.89% 75.56%	19.12% 0.97% 54.56%	43.96% 0.81% 42.61%	61.24% 0.89% 38.61%
nnUNet (Nat-Method v21) scratch	DONE 100ep	32.19% 0.82% 46.28%	31.60% 0.83% 56.63%	29.14% 0.83% 66.77%	15.44% 0.12% 56.63%	32.96% 0.82% 44.13%	56.23% 0.61% 36.75%
MONAI SwinViT (CVPR22) Tang SSL	DONE 4h24m 100ep	44.56% 0.85% 45.76%	29.27% 0.83% 76.19%	26.12% 0.83% 78.80%	15.46% 0.23% 57.61%	56.18% 0.84% 36.12%	61.60% 0.82% 37.82%
MONAI SwinViT scratch (CVPR22) scratch	DONE 4h21m 100ep	46.88% 0.85% 45.92%	32.84% 0.83% 71.17%	29.18% 0.83% 78.25%	11.48% 0.73% 59.83%	57.55% 0.85% 32.48%	61.62% 0.82% 34.31%
DynNet dual-enc (MICCAI 25) PET-CT dual-enc	DONE 100ep	45.00% 0.82% 36.76%	41.41% 0.83% 55.87%	36.25% 0.83% 64.74%	17.69% 0.12% 47.97%	49.19% 0.81% 33.83%	63.80% 0.69% 29.84%
DynNet (MICCAI 25) scratch	DONE 2h32m 100ep	43.96% 0.82% 38.89%	40.75% 0.89% 74.57%	41.23% 0.89% 73.66%	16.71% 0.12% 50.23%	44.83% 0.81% 39.39%	61.42% 0.68% 39.16%
SegAnyPET (ICCV 25) SegAnyPET-Lesion	DONE 6h14m 100ep	15.64% 0.86% 97.74%	8.48% 0.86% 99.67%	8.82% 0.86% 99.55%	25.88% 0.86% 90.59%	21.97% 0.86% 95.18%	44.68% 0.86% 72.37%
SegAnyPET scratch (ICCV 25) scratch	DONE 7h42m 100ep	7.95% 0.86% 99.24%	3.67% 0.86% 99.93%	3.83% 0.86% 99.78%	22.13% 0.81% 93.34%	19.16% 0.86% 97.28%	35.59% 0.49% 76.42%
PET/CT MAE (arXiv 26) PET/CT MAE SSL	DONE 4h23m 100ep	53.81% 0.86% 33.31%	39.33% 0.84% 57.84%	30.61% 0.83% 75.43%	15.98% 0.60% 31.24%	68.49% 0.86% 26.23%	73.20% 0.61% 26.62%
PET/CT MAE scratch (arXiv 26) scratch	DONE 4h41m 100ep	48.10% 0.85% 41.34%	38.23% 0.83% 62.46%	31.16% 0.83% 71.79%	15.62% 0.50% 45.44%	58.88% 0.85% 32.61%	66.30% 0.62% 26.59%
BIRTH scratch (AutoPET V26) scratch	DONE 100ep	RUNNING	32.47% 0.81% 69.80%	29.46% 0.89% 75.78%	-	-	-
BIRTH / bomingduo (AutoPET V26) Dataset619 MultiTalent	PENDING	-	-	-	-	-	-
YixinChen scratch (AutoPET V26) scratch	DONE 100ep	32.67% 0.82% 42.20%	28.39% 0.89% 73.87%	26.57% 0.89% 78.61%	-	-	-
YixinChen / cheniyixin (AutoPET V26) Dataset619 MultiTalent	PENDING	-	-	-	-	-	-

Figure 1: Mean TEST board (Dice / FP / FN). Nine-fold grids, ETA, and queue footers omitted. Incomplete rows remain pending/running.

4 DISCUSSION

Matching objectives to clinical concept drift means: (i) pretrain for transferable anatomical–metabolic structure on the source tracer; (ii) adapt with a protocol whose label budget is stated explicitly; (iii) evaluate with FP/FN under long-tailed empty/positive cases, not only mean Dice (Bercea et al., 2025b;a). Our board is designed so the core claim is falsifiable: if SSL never helped at fs50/fs10, or continued to dominate at fs5/fs0, the shot-dependent statement would be revised accordingly.

5 CONCLUSION

Under FDG→PSMA clinical concept drift, source-domain self-supervised representations accelerate PSMA adaptation at moderate few-shot budgets, but the benefit is not automatic in the extreme low-/zero-shot regime. Fair, aligned protocols and FN-aware metrics are required to locate that

boundary. We will freeze incomplete competition rows and add fold-wise intervals for the camera-ready.

REPRODUCIBILITY STATEMENT

Code, split JSONs, and the mean-metric board exporter: <https://github.com/yeebowang/fdg2psma-fewshot>. Raw volumes/checkpoints are not redistributed; launchers reproduce Fig. 1 and Table 1.

ETHICS STATEMENT

Research PET/CT under existing data-use agreements; no identifiable patient data in the public repo. Clinical use would need prospective validation.

AI USE STATEMENT

LLMs assisted with LaTeX editing. Design, training, evaluation, and numbers come from the authors' pipelines and were checked against board JSON.

REFERENCES

- Anonymous. DpDNet. In *MICCAI*, 2025a. Placeholder; replace with the official DpDNet citation.
- Anonymous. SegAnyPET: Towards interactive lesion segmentation in whole-body PET/CT. In *ICCV*, 2025b. Venue/year as used on our aligned board; update with final citation.
- Anonymous. 3D masked autoencoders for PET/CT representation learning. *arXiv preprint*, 2026. Placeholder for the PET/CT MAE preprint cited on the board.
- Cosmin I Bercea, Jun Li, Philipp Raffler, et al. NOVA: A benchmark for rare anomaly localization and clinical reasoning in brain MRI. In *NeurIPS Datasets and Benchmarks*, 2025a.
- Cosmin I Bercea, Benedikt Wiestler, Daniel Rueckert, and Julia A Schnabel. Evaluating normative representation learning in generative AI for robust anomaly detection in brain imaging. *Nature Communications*, 2025b.
- Yixin Chen et al. From FDG to PSMA: A hitchhiker's guide to multitracer, multicenter lesion segmentation in PET/CT imaging. In *AutoPET Challenge / related report*, 2025. Update with camera-ready citation.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *CVPR*, 2022.
- Fabian Isensee, Paul F Jaeger, Simon A A Kohl, Jens Petersen, and Klaus H Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.
- Yucheng Tang, Dong Yang, Wenqi Li, et al. Self-supervised pre-training of swin transformers for 3d medical image analysis. In *CVPR*, 2022.

A PROTOCOL POINTERS

See `run/` and `scripts/iclr2026-aligned-fdg-fs50-board.py`. Per-fold dictionaries remain in internal board JSON for auditing.